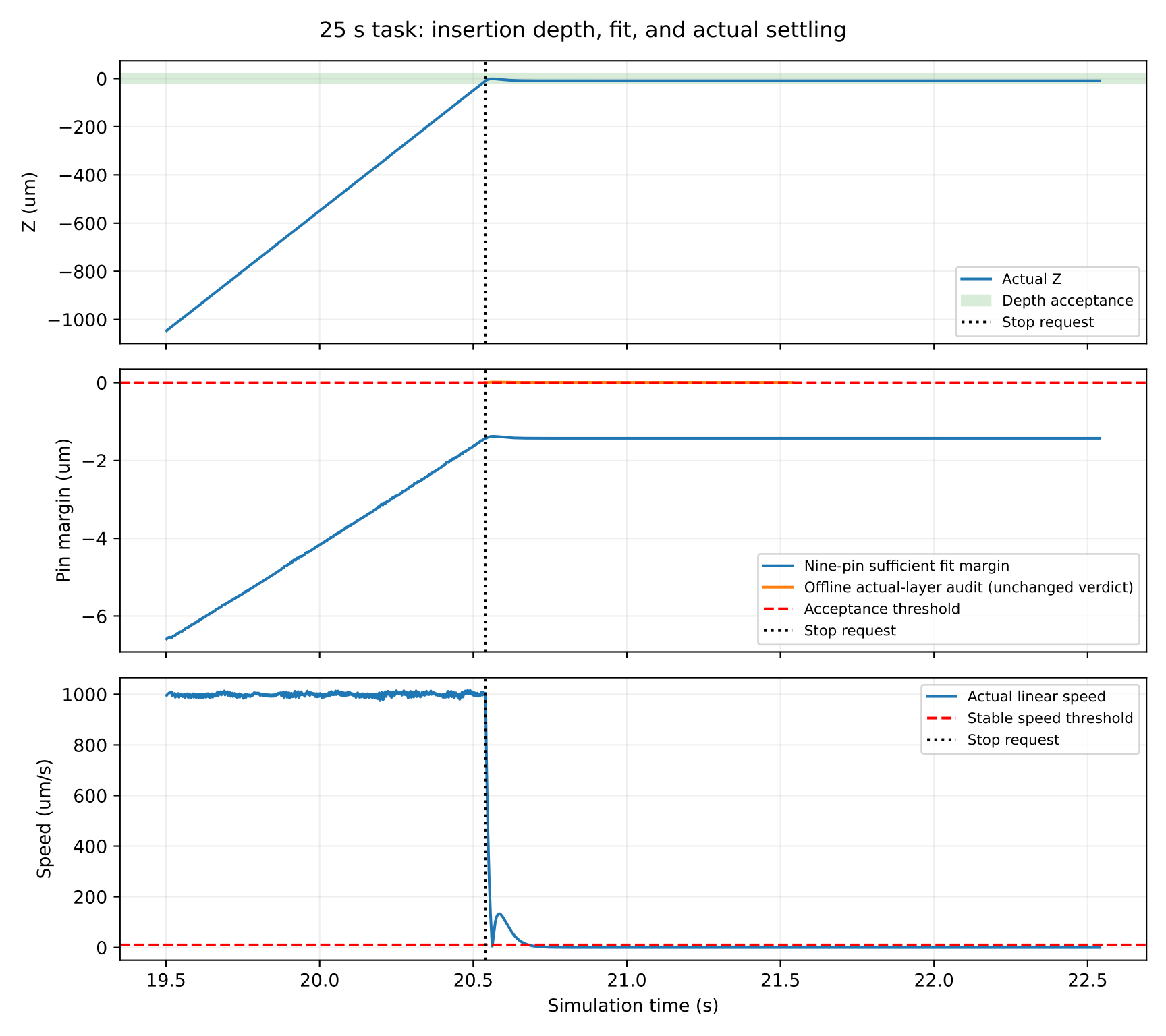
25 s task: insertion depth, fit, and actual settling
Z (um)
0
−200
−400
−600
−800
−1000
Actual Z
Depth acceptance
Stop request
Pin margin (um)
0
−2
−4
−6
Nine-pin sufficient fit margin
Offline actual-layer audit (unchanged verdict)
Acceptance threshold
Stop request
Speed (um/s)
1000
800
600
400
200
0
Actual linear speed
Stable speed threshold
Stop request
19.5
20.0
20.5
21.0
21.5
22.0
22.5
Simulation time (s)